Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Mickey Mouse Clubhouse
Project / Solution Name	Orbyte Collective
Name (Leader)	Leow Wei Xin
Email & Phone (Leader)	018-3907912 (leowweixin@gmail.com)
University Affiliation(s)	Universiti Malaya
Live Prototype URL	https://sdc-collective.vercel.app/
Code Repository URL (public / access granted to judges)	https://github.com/FlyingShuriken/collective

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Orbyte Collective assists small Malaysian businesses with repeat purchasing. A Next.js mobile web app supports English, Bahasa Melayu and Chinese. A Python FastAPI backend uses Supabase PostgreSQL, Auth and private Storage. Three AI readers propose invoice/stock records, supplier quotes/messages and product-specification matches. People review evidence and confirm records before they enter the ledger. Deterministic code calculates replenishment, delivered costs, quantity options, invoice differences and pool eligibility. Each owner approves bound purchasing terms; changed terms or an exceeded cap invalidate approval. Learned usage, delivery-fee and quote-accuracy rules require explicit owner or network-admin acceptance. Durable jobs handle reading, rule proposals, reorder notices and expiry. Tenant access controls and append-only events support privacy and auditability. Purchase orders and supplier messages are simulated; no payments move. The default demo uses synthetic record/replay responses and recorded BGE-M3 embeddings. Live model adapters and deployment configuration exist, but live accuracy and hosted readiness are not established by replay tests.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.x; Next.js 16.4.2; React/React DOM 19.3.0	Web interface, routing and UI rendering	Apache-2.0 (TypeScript); MIT (Next.js/React)
Tailwind CSS 4.x; shadcn 4.22.x; Radix UI 1.8.x; lucide-react 1.61.x	Styling, accessible UI primitives and icons	MIT; lucide icons: ISC (check bundled notices)
@supabase/supabase-js 2.118.0; @supabase/ssr 0.12.7	Database/Auth/Storage client and server-side sessions	MIT
class-variance-authority 0.7.x; cn 0.5.x; tw-animate-css 1.4.x	UI class composition and animations	Verify each installed package license before signing

Name & Version	Purpose in Project	License
Python 3.12.13; FastAPI 0.161.1; Pydantic 2.14.1; pydantic-settings 2.16.1	API, validation, configuration and deterministic core	PSF (Python); MIT (listed Python libraries)
Uvicorn 0.54.0; psycopg 3.3.6; HTTPX 0.28.1; PyJWT 2.15.1	ASGI serving, PostgreSQL access, HTTP and JWT verification	BSD-3-Clause; LGPL-3.0 (psycopg); BSD-3-Clause; MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Supabase Auth, Storage and PostgreSQL interfaces	Authentication, private source files and tenant-scoped records; local stack documented	Free
Google Gemini Developer API (configured adapter)	Invoice/stock reading, supplier-quote parsing, message drafting and shortlist checking; default demo uses replay	Free
BGE-M3 local/HTTPS embedding endpoint (optional)	Multilingual product-candidate retrieval; demo uses recorded local vectors	Local recorded inference

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development environment, reported by team	Official sponsor-provided access
Supabase	PostgreSQL, Auth and private Storage	Local Docker deployment; Free cloud plan
Vercel	Next.js deployment target	Free plan in southeast asia
Render	Docker API and background worker deployment target	Free plan

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
BGE-M3; gpustack/bge-m3-GGUF revision 2d48f1737679ad900d5c26c5aad5410e9c70fdca	BAAI; GPUStack quantized artifact	Pre-trained; local vectors recorded for synthetic matching cases	BGE-M3: MIT
gemini-3.8-flash	Google Gemini adapter configuration	Configured pre-trained model	Google service terms;
gemini-4-argon	Google Gemini adapter configuration	Configured escalation model;	Google service terms;

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Three-café, fee-warning and five-business history scenarios	Repository: fixtures/scenarios/	Synthetic	Project-authored	N
Invoice/PDF/image/text and gold-field fixtures	Repository: fixtures/sources/ and fixtures/gold/	Synthetic	Project-authored	N
Multilingual matching cases and recorded vectors	Repository: fixtures/matching/	Synthetic inputs; recorded pre-trained embeddings	Project fixtures plus upstream model terms	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Authored model replay records	Repository: fixtures/replay/ (generated as documented)	Synthetic; not measured live-model outputs	Project-authored	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✔️ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit	Development assistance	minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We review extracted values against their source, correct records before confirmation, and keep purchasing decisions in deterministic code. Tests cover calculations, permissions, approvals, data isolation and workflows. Owners approve exact message revisions and bound orders. Synthetic replay verifies workflow behaviour, not live-model accuracy.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✔️ Yes ☐ No

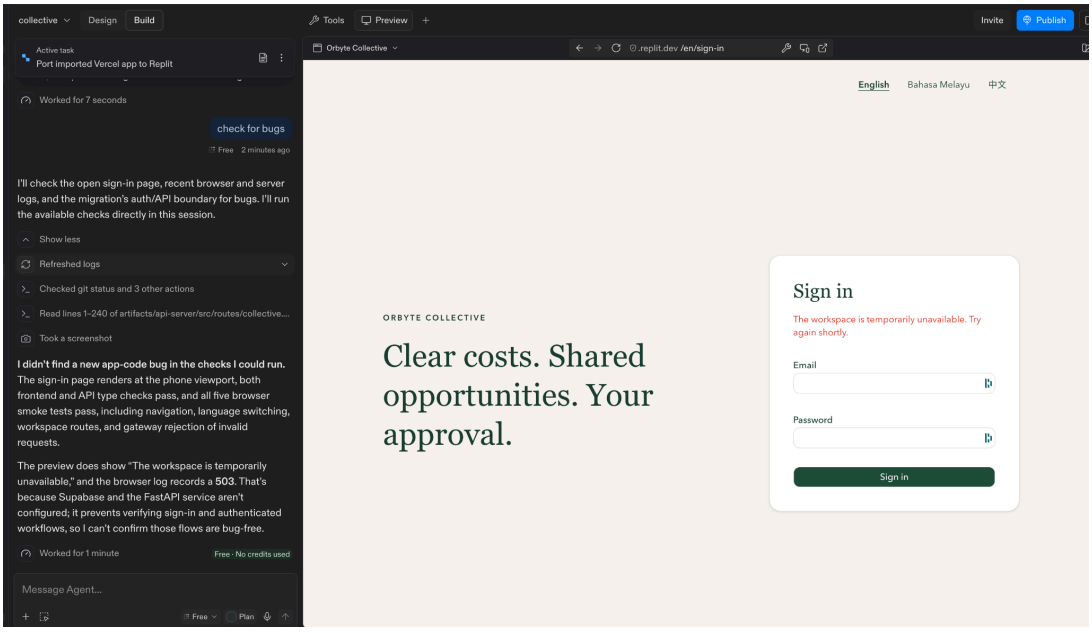
If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit development platform	Used during project development with official sponsor-provided access

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit supports agentic planning and development, we are using it to do bug exploring and fixing alongside with local development. Replit supported development but is replaceable. The repository documents local setup with Node.js, Python and the Supabase stack. Core purchasing calculations and approval rules do not require Replit. The AI gateway separates model reading from decisions and supports replay.



6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype stores information such as user accounts, business roles, supplier details, uploaded documents, purchasing and inventory records, quotations, approvals. All purchasing data used for the demo is simulated.

To protect this information, the system uses authentication, tenant-based access controls, PostgreSQL row-level security, private file storage, and role-based permissions. Business owners can review and remove sensitive information before documents are processed, and only the prepared text or images are sent to the configured AI services.

AI agents cannot independently release orders or send information to suppliers. Sensitive credentials are stored in environment configurations and are not included in this declaration.

Currently, there is no automatic data retention or deletion policy. Original files and audit records remain stored until an authorized person manages or removes them. Before the system is used with real customer data, a proper data retention policy and clear data-processing agreements with AI service providers must be established.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The system may misread prices, quantities, dates, or product details. To reduce errors, it checks source documents, validates calculations, flags uncertain results, and requires human approval for important decisions. It also checks product specifications to prevent incorrect matches.

AI services may fail or return invalid results, so the system uses limited retries and clear error messages. Current tests rely on simulated data, and accuracy testing with 30 real documents is still pending.

The prototype is designed for small-scale use, with orders and messages simulated. Production readiness, data retention, and device installation remain unverified. Two integration tests initially failed but passed when retried individually, and the cause is still unknown.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

(/) All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

(/) All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

(/) No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

(/) Any use of generative-AI tools for development has been disclosed and reviewed by the team.

(/) Any use of official SDC sponsor resources has been declared in Section 5.

(/) The prototype described here is accessible to judges through the links provided in Section 1.

(/) All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

25/10/2026